

Moore Example:

Entity moore_fsm is

Port

clk: in std_logic; reset: in std_logic;
input: in std_logic;
output: out std_logic;

);

end moore_fsm;

Architecture behavioral of moore_state is

TYPE state IS (S0, S1, S2);

SIGNAL Moore_state: state;

U_Moore: Process (clk, reset)

BEGIN

IF (reset = '1') THEN

moore_state <= S0;

ELSE IF (clk = '1' and clk'event) THEN

CASE moore_state IS

WHEN S0 =>

IF input = '1' THEN

moore_state <= S1;

ELSE

moore_state <= S0;

END IF;

WHEN S1 =>

IF input = '0' THEN

moore_state <= S2;

ELSE

moore_state <= S1;

END IF;

WHEN S2 =>

IF input = '0' THEN

moore_state <= S1;

END IF;

END CASE;

END IF;

END PROCESS;

Output <= '1' WHEN moore_state = S2 ELSE '0'

END behavioral;

testbench;

entity moore_fsm_tb is

end moore_fsm_tb;

Architecture behavior of moore_fsm_tb is

component moore_fsm

Port

clk: in std_logic;

reset: in std_logic;

input: in std_logic_vector (1 downto 0);

output: out std_logic_vector (1 downto 0);

);

end component;

SIGNAL clk_s: std_logic := '0';

SIGNAL reset_s: std_logic :=